Retail Business Performance & Profitability Analysis

Introduction

Retail businesses operate on thin and highly variable margins across product categories, regions, and seasons. Without visibility into which categories, sub-categories, and inventory practices are draining profit, businesses risk over-investing in low-margin, slow-moving stock while under-investing in high-margin opportunities. This project analyzes transactional retail data to uncover profit-draining categories, evaluate the relationship between inventory turnover and profitability, and identify seasonal product behavior — with the goal of producing clear, actionable recommendations for margin improvement.

Abstract

This analysis covers 6,000 retail transactions (5,821 after cleaning) spanning five categories — Electronics, Apparel, Home & Kitchen, Grocery, and Beauty & Personal Care — across five regions and five seasons. The dataset was cleaned and loaded into SQL for aggregation, then analyzed in Python to test the relationship between inventory holding time and profitability. Key findings show an overall profit margin of **11.66%** on **$26.87M** in revenue, with **Beauty & Personal Care (20.67%)** and **Apparel (17.84%)** as the strongest-margin categories, while **Grocery (2.59%)** and **Electronics (5.08%)** are the primary profit drains despite Electronics generating the highest raw revenue. Within every category, longer inventory holding times are consistently associated with lower margins (correlation range: **-0.54 to -0.63**), driven by markdown pricing on slow-moving stock. A dashboard and 10 supporting SQL queries were built to operationalize these findings for ongoing monitoring.

Tools Used

- **SQL (SQLite):** Data cleaning, profit margin aggregation by category/sub-category/region, inventory-bucket segmentation, window functions for ranking
- **Python (Pandas, NumPy):** Correlation analysis between inventory days and profitability, both overall and within category
- **Matplotlib/Seaborn:** 9-panel visual dashboard covering revenue, margin, regional, seasonal, and inventory-turnover views
- **CSV/SQLite:** Data storage and exchange format for all deliverables

Steps Involved in Building the Project

1. **Data Preparation:** Generated/imported a transactional retail dataset (6,000 rows) with category, sub-category, region, season, cost, price, quantity, and inventory-days fields; introduced realistic missing values to simulate real-world data quality issues.
2. **Data Cleaning (SQL):** Identified null records in `region` and `inventory_days` (2.4% of rows) and created a cleaned view excluding incomplete records, preserving the raw table for auditability.
3. **Profit Margin Analysis (SQL):** Aggregated revenue, cost, and profit by category and by category+sub-category to rank performance and isolate the lowest-margin sub-categories.
4. **Regional & Seasonal Analysis (SQL):** Broke down revenue and margin by region and by season×category to detect geographic or seasonal profitability shifts.
5. **Inventory-Profitability Correlation (Python):** Used Pandas to compute the correlation between `inventory_days` and `profit_margin_pct`, both in aggregate and within each category, to control for category-level margin differences (avoiding Simpson's paradox in the aggregate figure).
6. **Segmentation for Action (SQL):** Used a CTE to flag category/sub-category/region combinations with inventory days above 60 and margins below their own category average — these are the specific segments needing markdown or replenishment strategy changes.
7. **Dashboard Build:** Created a 9-panel dashboard (revenue by category, margin by category, profit share, regional revenue, seasonal trends, inventory-vs-margin scatter, turnover-bucket margins, lowest-margin sub-categories, revenue-vs-profit comparison).
8. **Recommendation Synthesis:** Translated the quantitative findings into prioritized, category-specific business actions.

Key Insights & Recommendations

- **Grocery and Electronics drain margin despite driving volume.** Grocery has the fastest turnover (avg. 3–20 days) but the thinnest margin (2.59%); Electronics generates the most revenue ($14.9M) but converts only 5.08% to profit. **Action:** renegotiate supplier costs or introduce margin-accretive bundling for these categories rather than competing purely on price.

- **Longer inventory holding time predicts lower margin within every category** (correlation -0.54 to -0.63), confirming that markdown pressure on aging stock is a consistent margin drain. **Action:** tighten reorder cycles for Apparel and Home & Kitchen sub-categories, which show the longest average holding times (75–80 days).
- **Ten specific category-region-subcategory segments** (mostly in Apparel — Women's Wear, Footwear, Men's Wear — plus Home & Kitchen Furniture) combine long holding times with below-category-average margins. **Action:** prioritize these for clearance pricing or reduced future purchase orders.
- **Beauty & Personal Care is the highest-margin category (20.67%)** despite modest revenue share. **Action:** expand catalog depth and marketing investment here, as it offers the best return per dollar of revenue.
- **Regional performance is fairly even** (11.4%–11.9% margin across all five regions), suggesting profitability issues are category-driven rather than geography-driven — retention/expansion strategy should be uniform across regions.

Conclusion

This analysis demonstrates that profitability in retail is shaped less by top-line revenue and more by the interaction between category margin structure and inventory discipline. Electronics and Grocery — the two largest revenue drivers — are simultaneously the two largest sources of margin erosion, while high-margin categories like Beauty & Personal Care remain under-leveraged. The consistent negative correlation between inventory days and margin across every category confirms that tighter inventory management is a lever available to all categories, not just a subset. The SQL queries and dashboard built in this project provide a repeatable framework the business can rerun each reporting period to track whether these interventions are improving overall profitability.